Luminosity

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Luminosity Estimation

Goal: Estimate the luminosity of data we have on disk with |z| < 10 cm.

Trigger Definitions

$$\text{Trigger 12} = \text{MBD N\&S} \geq 2,\ \text{vtx} < 10\ \text{cm (Narrow)}$$
$$\text{Trigger 14} = \text{MBD N\&S} \geq 2,\ \text{vtx} < 150\ \text{cm (Wide)}$$

$$\text{Raw[X]} = \text{Raw counts for Trigger X.}$$
$$\text{Live[X]} = \text{Live counts for Trigger X.}$$
$$\text{Scaled[X]} = \text{Scaled counts for Trigger X.}$$

Vertex Acceptance Correction

$$z_{\text{frac}} = \left(\frac{\text{Events with } |z| < 10\ \text{cm}}{\text{Total Events}}\right) \approx \frac{\text{Raw[12]}}{\text{Raw[14]}}$$

$$\sigma_{\text{narrow}} = \sigma_{\text{MBD}} \cdot z_{\text{frac}}$$

Assumptions

$$\sigma_{\text{inel}} = 6.8\ \text{barns}$$
$$\epsilon_{\text{eff}} = 0.93$$
$$\sigma_{\text{MBD}} = \sigma_{\text{inel}} \cdot \epsilon_{\text{eff}} = 6.324\ \text{barns}$$

Events

Scenario A: Trigger 14 (Wide) has NO prescale

- Let N = Scaled[14] * z_{frac}

Scenario B: Trigger 14 (Wide) HAS a prescale

- Let N = Scaled[12]

$$\text{N} = \begin{cases} \text{Scaled[14]} \cdot z_{\text{frac}} & \text{if Scaled[14] / Live[14]} > 0.99 \\ \text{Scaled[12]} & \text{Otherwise.} \end{cases}$$

Luminosity

$$\mathcal{L}_{\text{sampled}} = \frac{N}{\sigma_{\text{narrow}}}$$

Luminosity Estimation: Example Run 68144

DAQ Database

```
daq=> select runnumber, index, raw, live, scaled from gl1_scalers where runnumber = 68144 and index in (12,14);
 runnumber | index |    raw     |    live    |  scaled
-----------+-------+------------+------------+----------
     68144 |    12 | 26977085   | 24894381   | 24894381
     68144 |    14 | 52263333   | 48174105   | 40145087
```

Step 1: Calculate Vertex Acceptance Correction

$$z_{\text{frac}} \approx \frac{\text{Raw}[12]}{\text{Raw}[14]} = \frac{26977085}{52263333} \approx 0.516.$$

Step 2: Check Prescale Status

$$\frac{\text{Scaled}[14]}{\text{Live}[14]} = \frac{40145087}{48174105} \approx 0.83 < 0.99, \text{ so } N = \text{Scaled}[12].$$

Step 3: Estimate Sampled Narrow Luminosity

$$\mathcal{L} = \frac{N}{\sigma_{\text{narrow}}} = \frac{24894381}{(6.324 \text{ barns}) \cdot 0.516} \cdot \frac{1 \text{ barn}}{10^6 \mu\text{b}} \approx 7.63 \mu\text{b}^{-1}.$$

Run 3 Au+Au

- Analyzed 5 Runs: 68099, 68130, 68144, 76218, 78218

Luminosity (|z| < 10 cm) [μb^{-1}]

Run	Lumi
68099	7.58
68130	7.26
68144	7.63
76218	4.58
78218	11.31

$$\mathcal{L} = 38.4\ \mu\mathrm{b}^{-1}$$

Note: Runs are not fully transferred / produced so luminosity estimation using DB numbers is overestimating.

Run 3 Au+Au

- Plan to increase analysis to top 100 Runs <u>with most segments produced</u>

Luminosity (|z| < 10 cm) [μb^{-1}]

Run	Lumi	Run	Lumi	Run	Lumi	Run	Lumi	Run	Lumi
67580	5.49	68353	5.23	69007	9.29	69319	6.39	71852	8.47
67633	5.28	68389	5.52	69009	6.41	69323	9.15	71898	10.83
67695	7.22	68402	10.19	69010	5.46	69324	6.66	71899	7.96
67696	5.26	68406	6.29	69020	6.35	69328	8.57	71940	9.69
67740	8.00	68490	6.09	69031	7.22	69342	7.28	71952	10.19
67743	5.97	68504	6.72	69032	5.75	69797	10.83	71985	11.05
67746	5.38	68522	6.56	69049	6.46	71209	9.94	72019	10.82
67770	6.33	68584	0.10	69050	5.48	71220	9.59	72020	8.22
67785	7.92	68627	8.49	69073	7.24	71260	9.98	72383	10.92
67932	6.47	68629	6.24	69273	7.69	71263	7.48	72401	8.41
67934	5.51	68715	7.78	69274	6.00	71276	11.20	72422	6.89
68099	7.58	68903	6.13	69279	9.82	71329	10.94	72434	11.26
68102	5.55	68927	6.49	69282	6.43	71345	11.24	72462	11.23
68130	7.26	68945	9.12	69289	9.43	71378	11.47	72569	11.30
68134	5.47	68948	5.44	69302	10.49	71403	11.05	72710	11.04
68144	7.63	68963	10.31	69303	7.72	71416	9.78	72823	10.96
68145	6.62	68970	7.76	69309	9.19	71467	10.02	72830	10.92
68146	5.68	68973	6.11	69311	7.04	71814	7.97	72889	10.13
68152	8.05	68987	5.72	69312	5.96	71830	7.10	72921	10.70
68209	3.33	68999	8.31	69318	8.12	71844	10.11	72972	11.51

$$\mathcal{L} = 0.8\ \mathrm{nb}^{-1}$$

Note: Runs are not fully transferred / produced so luminosity estimation using DB numbers is overestimating.